GradPilot AI — Domain-Specialized Academic Compliance Agent

ET GenAI Hackathon 2026 | Problem Statement 5: Domain-Specialized AI Agents with Compliance Guardrails Team: Team Veritas | Neha Rajvanshi, GL BITM Greater Noida

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What is GradPilot AI?

GradPilot AI is a domain-specialized AI agent built for India's higher education institutions. It automates complex, multi-step academic workflows — curriculum compliance verification, student grievance resolution, exam planning, and career guidance — while staying strictly within UGC/AICTE regulatory guardrails at every step, with full auditability of every decision it makes.

The problem: India has 1,000+ universities and 40,000+ colleges. Most still manually verify curriculum compliance against UGC/AICTE norms, process student grievances through paper trails, and provide zero personalized academic guidance. GradPilot automates all of this with AI — transparently and compliantly.

Problem Statement Alignment (PS5)

PS5 Requirement	GradPilot Implementation
Domain-specific agent	Education domain — UGC/AICTE regulatory framework
Executes domain workflows	Curriculum compliance checks, grievance handling, timetable generation
Handles edge cases properly	Backlog students, mid-semester transfers, special exam accommodations
Regulatory & policy guardrails	All responses grounded in indexed UGC/AICTE PDFs via RAG
Auditable reasoning at each step	Every agent decision logged with source citation and confidence score

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Architecture

```
┌──────────────────────────────────────────────────────────────┐
│                GradPilot AI System                           │
│                                                              │
│ ┌──────────┐   ┌──────────────┐   ┌──────────────┐          │
│ │ Student  │   │ Institution  │   │ Admin/Faculty│          │
│ │ Portal   │   │ Dashboard    │   │ Panel        │          │
│ └────┬─────┘   └──────┬───────┘   └──────┬───────┘          │
│      └────────────────┼──────────────────┘                  │
│                       ▼                                      │
│              ┌──────────────────┐                            │
│              │ Query Processor  │                            │
│              │ (Intent Detection)│                           │
│              └──────────┬───────┘                            │
│                         │                                    │
│        ┌────────────────┼────────────────┐                   │
│        ▼                ▼                ▼                   │
│ ┌──────────────┐ ┌──────────────┐ ┌──────────────┐          │
│ │ Compliance   │ │ Academic     │ │ Career       │          │
│ │ Agent        │ │ Planning     │ │ Guidance     │          │
│ │ (UGC/AICTE)  │ │ Agent        │ │ Agent        │          │
│ └──────┬───────┘ └──────┬───────┘ └──────┬───────┘          │
│        └────────────────┼────────────────┘                   │
│                         ▼                                    │
│        ┌──────────────────────────┐                          │
│        │ RAG Pipeline             │                          │
```

```
│        │ Vector DB + LLM (Claude)│                │
│        └──────────┬──────────────┘                │
│                   ▼                               │
│        ┌──────────────────────────┐               │
│        │  Knowledge Base          │               │
│        │ UGC Regulations          │               │
│        │ AICTE Norms              │               │
│        │ University Syllabi       │               │
│        │ GATE/CAT/GRE Banks       │               │
│        └──────────────────────────┘               │
└──────────────────────────────────────────────────────┘
```

Agent Roles

1. Compliance Agent

- Checks any curriculum, exam schedule, or policy against UGC/AICTE norms
- Flags violations with specific regulation references
- Generates compliance-ready audit reports
- Guardrail: Will never suggest an action that violates a loaded regulation

2. Academic Planning Agent

- Generates personalized semester timetables (rule-based + AI)
- Handles edge cases: backlog subjects, elective clashes, makeup exams
- Provides syllabus-grounded doubt resolution via RAG

3. Career Guidance Agent

- Maps student profile → placement/higher studies pathways
- Grounds recommendations in real placement data (Kaggle datasets)
- Provides structured, citation-backed guidance — not hallucinated advice

How Agents Communicate

All agents share a common context object per session. The Query Processor routes intent to the right agent. Each agent response includes:

- source_chunks[] — retrieved document segments used
- regulation_refs[] — specific UGC/AICTE clause cited (if applicable)
- confidence_score — hallucination guard metric
- action_taken — logged for audit trail

Features

1. Curriculum Compliance Checker

Upload a course curriculum PDF → Agent verifies it against UGC/AICTE norms → Returns a compliance report with specific clause references and fix recommendations.

2. Personalized Semester Planner

Student inputs: branch, semester, backlog subjects, exam dates → Agent generates an adaptive weekly timetable with study priorities and revision windows.

3. AI Doubt Resolution (RAG)

Student asks a subject question → System retrieves relevant syllabus chunks + textbook content → LLM generates a structured, syllabus-grounded answer. Hallucination rate tracked per session.

4. Student Grievance Workflow Agent

Student submits grievance → Agent classifies it (academic/administrative/exam) → Routes to correct resolution pathway → Tracks status → Escalates stalls automatically.

5. Career Intelligence Module

Student inputs CGPA, skills, interests → Agent retrieves placement benchmarks → Generates a ranked list of career pathways with preparation roadmaps.

Impact Model

See IMPACT_MODEL.md for full quantified analysis.

Summary:

- 40,000+ colleges in India with zero automated compliance tooling
- Manual compliance audit: ~120 person-hours per institution per year → GradPilot reduces this to ~4 hours (97% reduction)
- Student grievance resolution time: 15–30 days manual → 2–4 days with GradPilot
- Addressable market: ₹2,400 Cr/year (B2B SaaS licensing at ₹6L/institution/year × 40,000 colleges)

Tech Stack

Layer	Technology
Frontend	React 18, TailwindCSS
Backend	Node.js, Express
AI/LLM	Claude API (claude-sonnet-4-20250514)
RAG Pipeline	LangChain.js + Pinecone (vector DB)
Document Parsing	pdf-parse, mammoth
Data Sources	UGC website PDFs, AICTE norms, public syllabi
Auth	JWT
Deployment	Vercel (frontend) + Render (backend)

Setup Instructions

Prerequisites

- Node.js v18+
- npm or yarn
- Anthropic API key
- Pinecone API key (free tier works)

Installation

```
# Clone the repository
git clone https://github.com/YOUR_USERNAME/gradpilot-ai.git
cd gradpilot-ai

# Install dependencies
npm install

# Set up environment variables
cp .env.example .env
# Edit .env and add your API keys:
# ANTHROPIC_API_KEY=your_key_here
# PINECONE_API_KEY=your_key_here
# PINECONE_INDEX=gradpilot-index

# Seed the vector database with knowledge base documents
npm run seed-kb

# Start development server
npm run dev
```

Environment Variables

```
ANTHROPIC_API_KEY=
PINECONE_API_KEY=
PINECONE_ENVIRONMENT=
PINECONE_INDEX=
JWT_SECRET=
```

PORT=3001

Project Structure

```
gradpilot-ai/
├── client/              # React frontend
|   ├── src/
|   |   ├── components/     # UI components
|   |   ├── pages/          # Route pages
|   |   └── hooks/          # Custom hooks
├── server/              # Node.js backend
|   ├── agents/
|   |   ├── complianceAgent.js
|   |   ├── planningAgent.js
|   |   └── careerAgent.js
|   ├── rag/
|   |   ├── vectorStore.js   # Pinecone integration
|   |   ├── retriever.js     # Document retrieval
|   |   └── pipeline.js      # RAG orchestration
|   ├── routes/            # API routes
|   └── middleware/        # Auth, logging, audit trail
├── knowledge-base/        # Source documents
|   ├── ugc-regulations/
|   ├── aicte-norms/
|   └── syllabi/
├── scripts/
```

```
|   └── seed-kb.js          # Vector DB seeding script
├── IMPACT_MODEL.md
├── ARCHITECTURE.md
└── README.md
```

Evaluation Criteria Mapping

Judging Criterion	Our Approach
Domain expertise depth	RAG grounded in official UGC/AICTE PDF documents — not generic LLM knowledge
Compliance & guardrail enforcement	Every response includes regulation_refs[]; agent refuses actions violating loaded norms
Edge case handling	Backlog students, mid-semester transfers, special exam accommodations all handled
Full task completion	End-to-end workflows: grievance intake → classification → resolution → closure
Auditability	Every agent decision logged with source chunk, confidence score, and action taken

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Demo

Demo flow:

1. Upload a sample curriculum PDF → watch Compliance Agent flag UGC violations in real time
2. Student logs in → generates personalized timetable for Semester 5 with 2 backlogs
3. Ask a subject doubt → see RAG retrieve syllabus chunk + generate grounded answer
4. Submit a grievance → watch the agent classify, route, and track it

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Team

Name	Role	Institution
Neha Rajvanshi	Full Stack + AI	GL BITM, Greater Noida

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